



Delanyo Kwame Bensah Kulevome

PH.D. STUDENT (博士生) · UESTC (电子科技大学)

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Summary (总结)

I am currently a PhD researcher in Information and Communication Engineering at the Yangtze Delta Region Institute (Huzhou). 10+ years academic and professional experience specializing in solving engineering problems using machine learning and deep learning. Self-motivated and interested in devising better problem-solving methods for challenging tasks, and learning new technologies and tools when the need arises.

Education (教育)

University of Electronic Science and Technology of China (UESTC)

Ph.D. in Information and Communication Engineering
GPA: 3.71/4.00

Chengdu, P. R. China
Sep. 2019 - present

University of Electronic Science and Technology of China (UESTC)

MEng. in Electronic Science and Technology
GPA: 3.99/4.00

Chengdu, P. R. China
Sep. 2017 - Jul. 2019

Accra Institute of Technology (AIT)

BEng. in Electrical/Electronic Engineering
GPA: 4.01/4.30 (summa cum laude)

Accra, Ghana
Oct. 2014 - May. 2017

Ho Polytechnic

HND. in Electrical/Electronic Engineering
GPA: 4.14/5.00

Ho, Ghana
Jan. 2007 - Jul. 2009

Research Interests (研究意向)

- Prognostics and Health Management (PHM) of Engineering Systems
- Deep Learning
- Transfer Learning
- Signal Processing
- Physics-Informed Neural Networks
- Condition Monitoring and Fault Diagnosis

Research Experience (研究经验)

ACADEMIC

- Academic researcher at the yangtze delta research institute of UESTC, (Aug. 2021 - present) Huzhou
- Fault diagnostics in a phased array system, (Sep. 2020 - present) Huzhou
- Roller bearing fault diagnosis and classification, (Sep. 2020 - present) Huzhou
- Classification of COVID-19 cases using deep learning models, (Sep. 2020 - Nov. 2020) UESTC, Chengdu
- Predicting the state of health of electrolytic capacitors, (Sep. 2019 - Dec. 2020) UESTC, Chengdu
- Predicting the remaining useful life (RUL) of lithium ion batteries, (Aug. 2018) UESTC, Chengdu

- Measuring and analyzing the RF field strength of a newly installed digital TV transmitter

Nationwide,
Ghana

PERSONAL

- Smart home automation system using Arduino
- Image recognition system using Raspberry pi 4
- Autonomous vehicle design, Coursera project
- Fault detection and classification in industrial machines using machine learning, MATLAB training project
- Intelligent water billing system using Arduino
- End of course projects on Udemy, Coursera, Pluralsight, and Udacity

Research Publications (研究出版物)

1. **Kulevome, D. B. K.**, Hong, Wang, & Xuegang, Wang. (2023). A Phased array system diagnostics and health management based on a dual-stream DCNN. *Physical Communication Journal*. (Under review)
2. **Kulevome, D. B. K.**, Wang, H., Wang, X., Kumar, R., & Cobbinah, B. (2023). Phased array antenna diagnosis from amplitude-only data using parallel deep learning models. *Journal of Applied Remote Sensing*, 17(1), 017502.
3. **Kulevome, D. B. K.**, Hong, Wang, & Xuegang, Wang. (2022). Rolling bearing fault diagnostics based on improved data augmentation and ConvNet. *Journal of Systems Engineering and Electronics*. (Accepted).
4. **Kulevome, D. B. K.**, Hong, Wang, & Xuegang, Wang. (2022). Deep neural network based classification of rolling element bearings and health degradation through comprehensive vibration signal analysis. *Journal of Systems Engineering and Electronics*, 33(1), 233-246.
5. Ayekai, B. J., Wenyu, C., Hailemichael, M. T., Fiasam, L. D., Kwaku, A. V., ... **Kulevome, D.**, & Mawuli, C. B. (2022). Federated Lung Cancer Prediction Using Histopathological Medical Images. In 2022 19th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP) (pp. 1-6). IEEE.
6. **Kulevome, D. K. B.**, Hong, W., Wang, X., Cobbinah, B. M., Agbley, B. L. Y., & Safder, Q. (2021). System Diagnosis Framework for Sustaining the Operational Fidelity of a Radar System. In 2021 18th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP) (pp. 640-644). IEEE.
7. Agbley, B. L. Y., Li, J., Haq, A. U., Bankas, E. K., Ahmad, S., Agyemang, I. O., **Kulevome, D.**, ... & Latipova, S. (2021). Multi-modal Melanoma Detection with Federated Learning. In 2021 18th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP) (pp. 238-244). IEEE.
8. **Kulevome, D. K.**, Wang, H., & Wang, X. (2021). A Bidirectional LSTM-Based Prognostication of Electrolytic Capacitor. *Progress In Electromagnetics Research C*, 109, 139-152.
9. Agbley, B. L. Y., Li, J., Haq, A., Cobbinah, B., **Kulevome, D.**, Agbefe, P. A., & Eleeza, B. (2020). Wavelet-based cough signal decomposition for multimodal classification. In 2020 17th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP) (pp. 5-9). IEEE.
10. Mawuli, E. S., Wu, Y., **Kulevome, D. K. B.**, Qingfeng, Z., Zhao, C., Liu, H., ... & Kai, K. (2020). Distributed Characterization of On-Chip Spiral Inductors for Millimeter-Wave Frequencies. In 2020 IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization (NEMO) (pp. 1-4). IEEE.

Patents (专利)

1. 王洪, 李磊, **Delanyo Kwame Bensah Kulevome**, 吴龙泉, 张昊坤, 金焱. (2022). 据增强和 CNN 的滚动轴承故障诊断方法. (已提交)。
2. 王洪, **Delanyo Kwame Bensah Kulevome**, 李磊, 吴龙泉, 张昊坤, 郭志成. (2022). DCNN 的相控阵雷达阵元故障诊断方法. (已提交)。

Work Experience (工作经验)

Freelance

Private Tutor

Ghana

Nov. 2016 – Jul. 2017

- Tutored junior high school (JHS) students in English, Mathematics, Science and other related subjects.
- Assisted students in completing assignments and projects.

SUBSEA 7

ROV Pilot Technician

Various locations worldwide
(Norway, Mexico, Angola, etc.)

Oct. 2011 – Jun. 2016

- Worked on various projects in Africa, Europe, and the Americas as a Remotely Operated Vehicles (ROV) pilot technician for clients such as Total, Exxon Mobil, and Statoil.
- Assisted in piloting and co-piloting an ROV on SUBSEA 7 vessels during subsea construction projects in various locations worldwide.
- Tested and installed Sonar, Camera, Laser, etc and other job specific devices on the ROV.
- Carried out dive checks before and after each subsea operation.
- Attended toolbox talks and task risk assessment meetings and completed necessary safety documentations prior to undertaking maintenance or repair works on the ROV system.

Ghana Broadcasting Corporation (GBC)

Assistant Technician Engineer

Ghana

Sep. 2009 – Oct. 2011

- Conducted research in the planning and installation laboratory and undertook reconfiguration of FM / TV transmitters and transposers.
- Developed alternative solutions to problems through research.
- Assisted in the installation, repair and maintenance of transmitters and transposers at various locations nationwide.
- Trained interns on the basic operations of a TV/Radio transmitter.

M & G Pharmaceuticals Company

Factory hand

Ghana

Feb. 2006 – Dec. 2006

- Blister machine operator.
- Assembled and disassembled relevant parts of the blister machine in preparation for a new product.
- Preprocessed and packaged pharmaceutical products.

Ministry of Defense (1 Signal Regiment)

Technician (National service person)

Ghana

Oct. 2004 – Jul. 2005

- Worked with other national service personnel to complete daily assigned projects.
- Assisted in troubleshooting and repairing of computers.
- Assisted in the installation of VHF/HF transceivers.

Internships (实习)

LEEFRANCO Electric Works

Intern

Accra, Ghana

Aug. 2008 – Oct. 2008

- Assisted in the troubleshooting of 11kV high tension supply and installation of 500kVA 11/.433kV transformer.
- Assisted in the earthing of substation equipment.

Ministry of Defense (1 Signal Regiment)

Intern

Ghana

Sep. 2007 – Nov. 2007

- Set up and operated public address systems at state and military functions.
- Assisted in the installation of VHF/HF transceivers.
- Gained and applied practical skills in the repair of computers, communication and test equipment.

Honors and Awards (荣誉奖项)

Domestic (Academic) (学术类奖项)

2021	1st Place, Academic achievement award in the school of information and communication eng., UESTC	Chengdu, China
2020	1st Place, Academic achievement award in the school of information and communication eng., UESTC	Chengdu, China
2019	Chinese University Scholarship (Full), Scholarship awarded for Ph.D. program in UESTC	Chengdu, China
2018	2nd Place, Academic achievement award in the school of information and communication eng., UESTC	Chengdu, China
2017	Chinese University Scholarship (Partial), Scholarship awarded for postgraduate taught program in UESTC	Chengdu, China

Domestic (Extracurricular) (其他奖项)

2021	3rd Place, Excellent performance award in the school of information and communication eng., UESTC	Chengdu, China
2021	2nd Place (Badminton), International students winter games, UESTC	Chengdu, China
2021	2nd Place (Lawn tennis), International students winter games, UESTC	Chengdu, China
2020	1st Place (Badminton), International students autumn games, UESTC	Chengdu, China
2018	2nd Place, Excellent performance award in the school of information and communication eng., UESTC	Chengdu, China

International (国际奖项)

2015	Safety Award, Award for enabling a safe working environment	subsea7, UK
2014	Safety Award, Award for enabling a safe working environment	subsea7, UK

Professional Membership (专业协会会员)

• Institute of Electrical and Electronics Engineers (IEEE)	Graduate Student Member
– IEEE Signal Processing Society Membership	
– IEEE Reliability Society Membership	
– IEEE Computer Society Membership	
– IEEE Young Professionals	
• National Society of Black Engineers (NSBE)	Member
• PHM society	Member
• Black in AI	Member

Professional Activities

• Optical Engineering SPIE	Journal External Reviewer
• The 5th International Conference on Mechanical, Electric & Industrial Engineering	External Reviewer
• Certified Peer Reviewer	Elsevier Researcher Academy

Skills (技能)

Computational skills	Programming: MATLAB; Python Frameworks: TensorFlow; Keras; PyTorch Certifications: TensorFlow Certification (Ongoing); Google Data Analytics (Ongoing)
Typesetting	MS Office Suite (Word, Excel, PowerPoint); LaTeX
Languages	English (Native); Chinese (Intermediate)
Personality	Hardworking; Self - motivated; Organized; Fast learner and eager to learn new things Excellent analytical skills; Good interpersonal skills; Good written and oral communication skills

Hobbies and interests (爱好兴趣)

Reading; Technology; Sports; Listening to music; Traveling

Resource links (证书)

Academic certificates	Higher National Diploma (HND) , Bachelor of Eng. (BEng) , Master of Eng. (MEng) , 硕士证书
Training certificates	Coursera & Udacity , Udemy
Awards and Honors	Domestic and International
Language	汉语水平考试 (HSK 3 certificate) - (Ongoing)
Others	Others